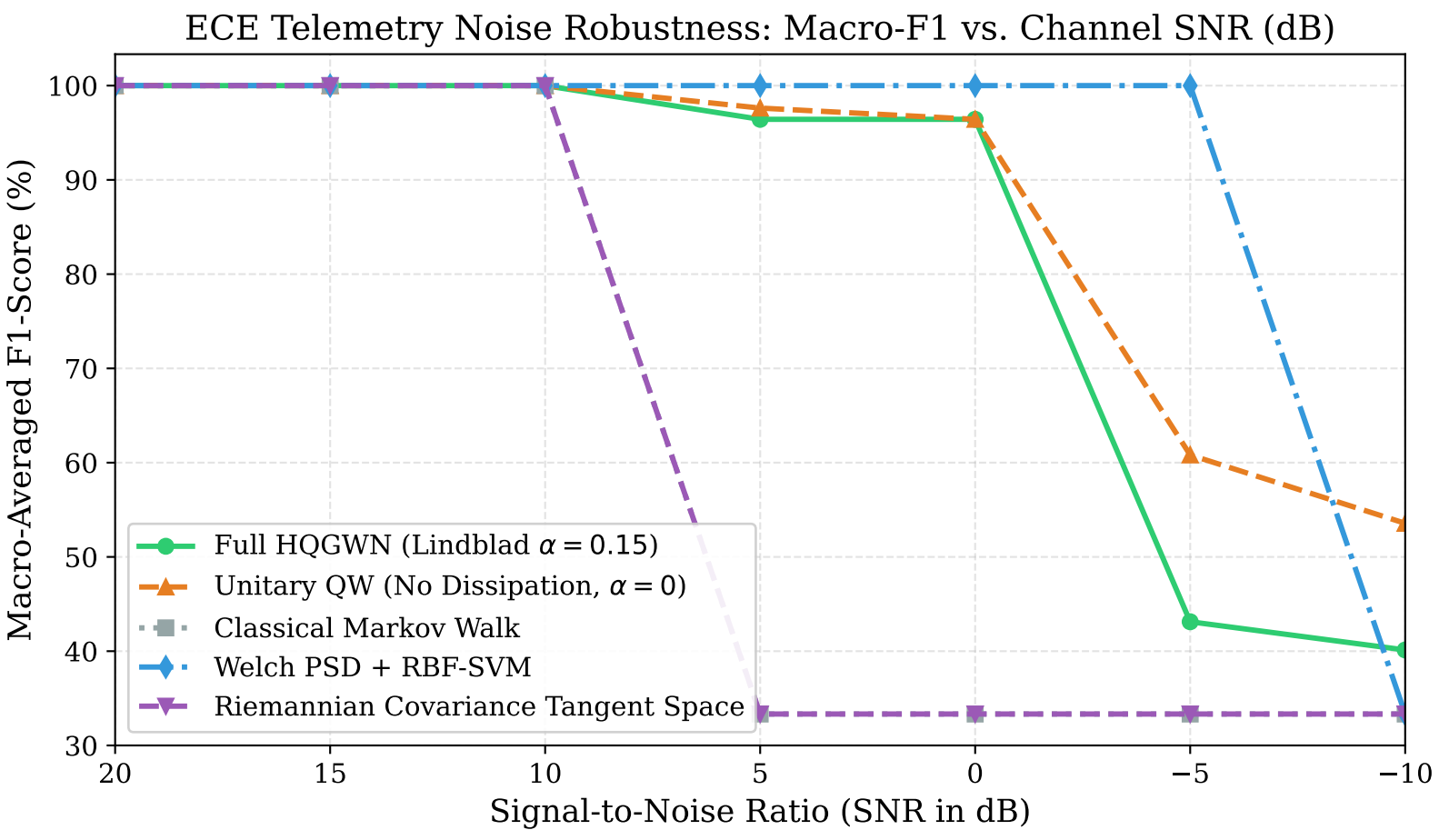
ECE Telemetry Noise Robustness: Macro-F1 vs. Channel SNR (dB)
Macro-Averaged F1-Score (%)
100
90
80
70
60
50
40
30
20
15
10
5
0
−5
−10
Signal-to-Noise Ratio (SNR in dB)
Full HQGWN (Lindblad $\alpha = 0.15$)
Unitary QW (No Dissipation, $\alpha = 0$)
Classical Markov Walk
Welch PSD + RBF-SVM
Riemannian Covariance Tangent Space